

Nerve cells must maintain the composition of their myelin sheaths to function properly. Galactocerebroside (GC), shown in Figure 1, is a hydrolyzable lipid and an important component of the myelin sheath.

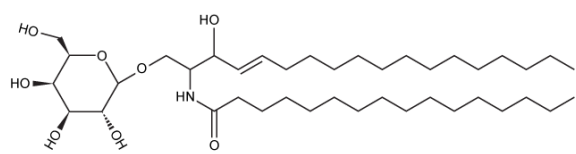


Figure 1 Structure of the myelin sheath component galactocerebroside

Homeostasis of GC is maintained by its dynamic synthesis and degradation, as shown in Figure 2. GC is synthesized by the enzyme ceramide galactosyltransferase (CGT) using UDP-galactose (UDP-Gal) and ceramide (Cer) as substrates. It is degraded by the enzyme galactocerebrosidase (GalC), which hydrolyzes the glycosidic bond to release galactose (Gal) and ceramide. A deficiency in GalC activity leads to a buildup of GC, and prior research has connected this buildup with neurodegeneration.

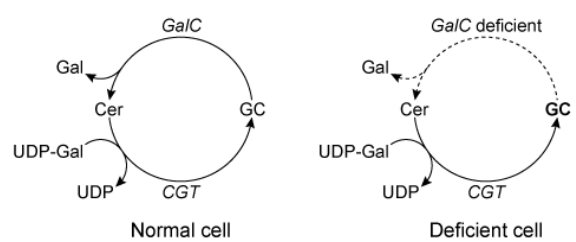


Figure 2 Schematic of CGT and GalC activity in healthy and GalC deficient cells

Over 100 inactivating GalC mutations have been identified. Five of the most common mutations are listed in Table 1. The changes to both DNA and amino acid sequence are shown.

Table 1 Five Common Galactocerebrosidase Mutations

Mutation type	Change in nucleotide sequence	Change in amino acid sequence
Missense	T→C	I250T
Deletion	TAA deleted	K139 deleted
Missense	A→C	D610A
Missense	A→T	D187V
Nonsense	G→A	W132X

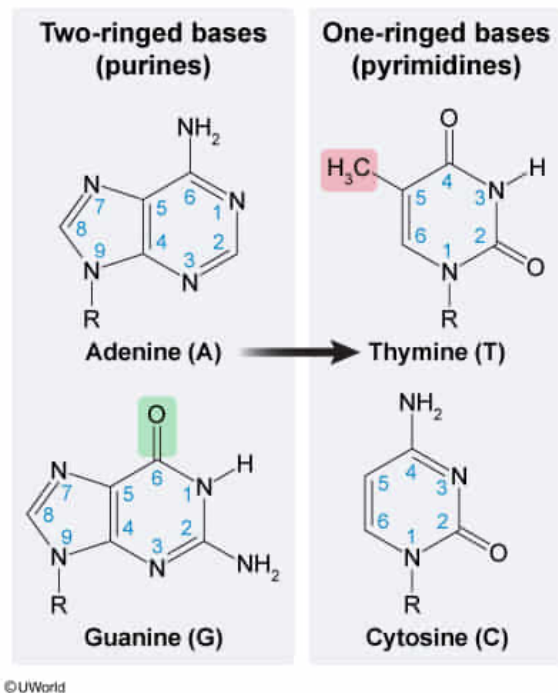
Based on Table 1, the D187V form of GalC resulted from what type of nucleotide change?

- ☐ A. A pyrimidine was replaced by a different pyrimidine [2%]
- ☐ B. A pyrimidine was replaced by a purine [4%]
- ☒ C. A purine was replaced by a pyrimidine [90%]
- ☐ D. A purine was replaced by a different purine [2%]

Correct

91% answered correctly

Explanation:



Nucleotides are classified based on the aromatic compounds from which their nitrogen bases are derived (ie, pyrimidine or purine). The compound **pyrimidine** is composed of a **single aromatic ring** containing two nitrogen atoms, which is the basis for **thymine** (or uracil in RNA) and **cytosine**. The compound **purine** is composed of **two aromatic rings**, essentially pyrimidine fused to a five-member **imidazole** ring, which is the basis for **adenine and guanine**. The mnemonic "PURE As Gold" is helpful in remembering that purines are associated with A and G nucleotides; the remaining nucleotides are pyrimidines.

According to Table 1, the D187V variant of GalC arises due to an A-to-T mutation in the DNA sequence. Adenine is a purine nucleotide, and thymine is a pyrimidine. Therefore, the D187V form of GalC results from **replacement** of a **purine (A)** by a **pyrimidine (T)**.

(Choice A) Adenine is a purine, not a pyrimidine.

(Choice B) Replacement of a pyrimidine by a purine would be correct if the mutation were thymine to adenine instead of adenine to thymine.

(Choice D) Thymine is a pyrimidine, not a purine.

Educational objective:

Nucleotides are classified as purines and pyrimidines based on the molecule from which they are derived. Pyrimidines contain a single aromatic ring (eg, cytosine, thymine, uracil). Purines contain two fused aromatic rings (eg, adenine, guanine).

Subject.

Biochemistry

Carbohydrates, Nucleotides, and Lipids

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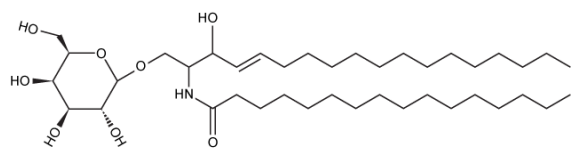


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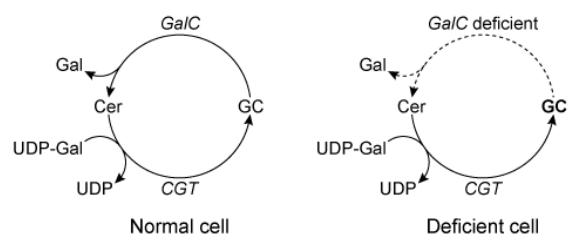


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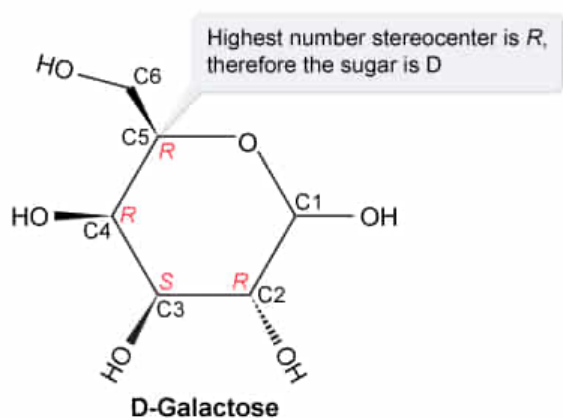
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What is the stereochemical configuration of the galactose component of GC?

- ☐ A. L [22%]
- ☒ B. D [49%]
- ☐ C. R [18%]
- ☐ D. S [10%]

Explanation:



Note: Stereochemistry of C1 (anomeric carbon) is not shown because the anomeric carbon can exist in either conformation.

Note: Stereochemistry of C4 is S-configuration in linear conformations of galactose.

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Molecules can have one or more **stereocenters** (an atom bonded to four unique substituents), each of which can adopt one of two stereochemical configurations with distinct properties. Every stereocenter within a molecule is given an **R or S designation** based on the **relative positions** of the substituents and their **priority ranking**. Priority is determined by the atomic number of each atom to which the stereocenter is bonded, with higher number given higher priority. If two or more of the bonds are to atoms of the same atomic number, then the atoms to which *they* are bonded are assessed until a difference is found.

Except for **dihydroxyacetone**, all carbohydrates have at least one stereocenter. Most monosaccharides (single sugar units) contain multiple stereocenters and cannot be designated as *R* or *S* because *R* and *S* configurations refer only to *individual* stereocenters. Instead, each carbon is given a number starting at one end of the chain. The anomeric carbon (two bonds to oxygen) is given the lowest number possible (ie, carbon 1 for aldoses, carbon 2 for ketoses), and the carbohydrate is then given an **L or D designation** based on the configuration of the stereocenter with the highest number (ie, the stereocenter farthest from the anomeric carbon). An *R* configuration at this position corresponds to a D-sugar, and an *S* configuration corresponds to an L-sugar. Biological structures such as the myelin sheath contain L-amino acids and **D-sugars almost exclusively**. The galactose component of GC shown in Figure 1 has an **R configuration at carbon 5**, making it a D-sugar.

(Choice A) L-Galactose is not used in biological structures. Figure 1 shows D-galactose.

(Choices C and D) *R* and *S* designations are used to classify individual stereocenters within a molecule. D-Galactose has both *R* and *S* stereocenters, but the molecule as a whole cannot be classified as either *R* or *S*.

Educational objective:

Stereocenters are designated as having *R* or *S* configurations based on the arrangement of their substituents and the priority ranking of each. Carbohydrates with multiple

stereocenters are given an L or D designation based on the configuration of the highest-numbered stereocenter. Almost all carbohydrates found in nature are in the D configuration.

Time spent:0 sec

Subject:
Biochemistry

QID:401931

System:
Carbohydrates, Nucleotides, and Lipids

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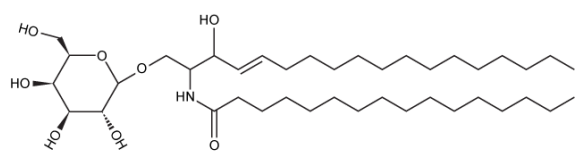


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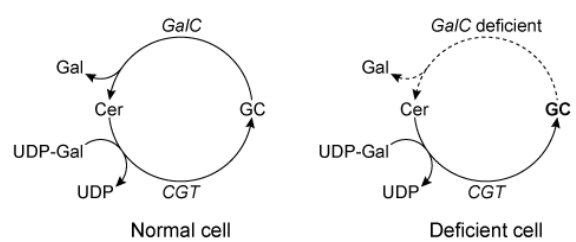


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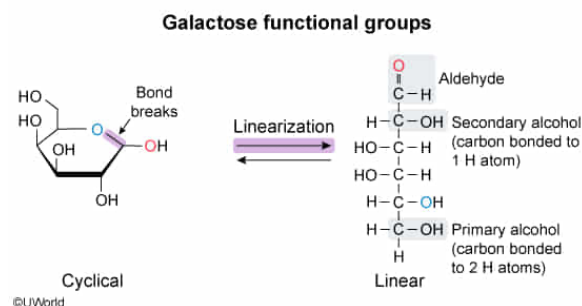
After galactose is released from GC and linearized, galactose contains all the following functional groups EXCEPT:

- ☐ A. an aldehyde. [22%]
- ☐ B. a primary alcohol. [5%]
- ☒ C. a ketone. [66%]
- ☐ D. a secondary alcohol. [6%]

Correct

66% answered correctly

Explanation:



Monosaccharides are chains of three or more carbon atoms with the general formula $C_nH_{2n}O_n$. Those with five or more carbons generally exist in equilibrium between **cyclic and linear forms**. In their linear forms, all monosaccharides contain one carbonyl group; all other carbons in the chain are alcohols because they are bonded to OH groups. If the carbonyl is on one of the *ends* of the chain, it is an **aldehyde**. Monosaccharides with an aldehyde group are known as **aldoses**. Carbonyls *within* the carbon chain are **ketones**, and monosaccharides with a ketone functional group are known as **ketoses**.

Galactose is an aldose, and as such it contains an **aldehyde (Choice A)** when released from GC and linearized. All the other carbons in galactose are alcohols. The alcohols in the middle of the chain have one C–H bond, making them **secondary alcohols (Choice D)**. The alcohol at the end of the chain has two C–H bonds, making it a **primary alcohol (Choice B)**. Galactose is **not a ketose**, so it does *not* contain a **ketone group**.

Educational objective:

In their linear form, monosaccharides contain multiple alcohol groups and one carbonyl group. For aldoses, the carbonyl is an aldehyde whereas for ketoses, the carbonyl is a ketone. All monosaccharides contain at least one primary alcohol, and most contain at least one secondary alcohol.

Time spent: 0 sec

Subject:
Biochemistry

QID: 401932

System:
Carbohydrates, Nucleotides, and Lipids

[illegible]

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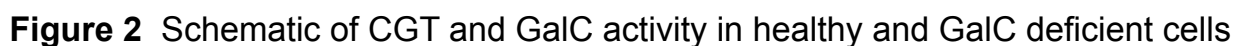


Table 1 Five Common Galactocerebrosidase Mutations

Based on the information in the passage, which bond is cleaved by GalC?

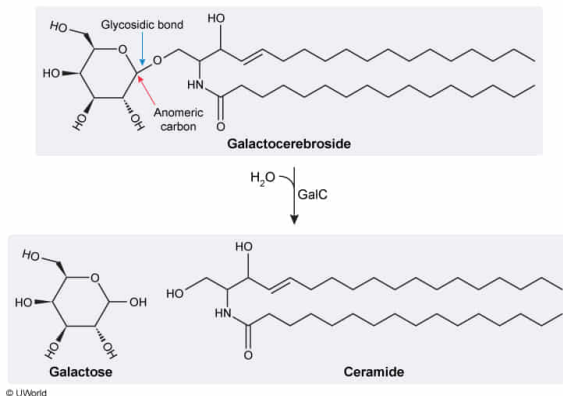
- 微信Mai_ange

☐ D. The bond between carbon 4 of galactose and an oxygen atom of ceramide [3%]

Correct

83% answered correctly

Explanation:



All monosaccharides have one carbon atom that has two bonds to oxygen, known as the **anomeric carbon**. The chain-terminating carbon that is closest to the anomeric carbon is designated as position 1, and all other carbons are numbered sequentially. In their linear form, **aldoses** such as **galactose** contain an **aldehyde group**. Because **aldehydes** are always at the **end of a carbon chain** (chain-terminating groups), the anomeric carbon of an aldose is always **carbon 1**. Ketoses contain **ketone groups**, which are not chain-terminating. Although a ketose could theoretically have a carbonyl at any non-chain-terminating carbon, biologically important **ketoses** (eg, **fructose**) have their anomeric carbon at **carbon 2**.

A **glycosidic bond** links the anomeric carbon of a carbohydrate to another molecule such as a lipid, a protein, a nitrogen base, or another carbohydrate. According to the passage, GalC **hydrolyzes a glycosidic bond** in GC to release the carbohydrate galactose and the lipid ceramide. Therefore, GalC cleaves the bond between carbon 1 of galactose and an oxygen atom of ceramide.

(Choice B) The bond between carbon 1 of galactose and the oxygen atom in the galactose ring links galactose to itself, not to another molecule. It is not the glycosidic bond.

(Choice C) The oxygen atom in the ring of galactose does not link galactose to another molecule. In addition, carbon 4 is not bonded to the ring oxygen.

(Choice D) Carbon 4 is not the anomeric carbon and does not participate in the glycosidic bond.

Educational objective:

Glycosidic bonds link the anomeric carbon of a carbohydrate to another molecule, such as a lipid, a protein, a nitrogen base, or another carbohydrate. The anomeric carbon has two bonds to oxygen: For aldoses, the anomeric carbon is carbon 1, and for biologically

important ketoses, it is carbon 2.

Time spent:0 sec

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Biochemistry

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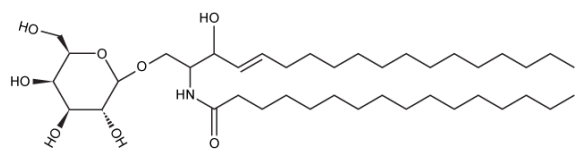


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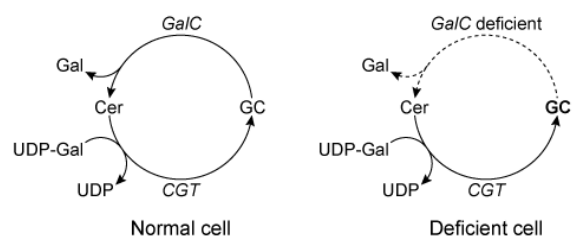


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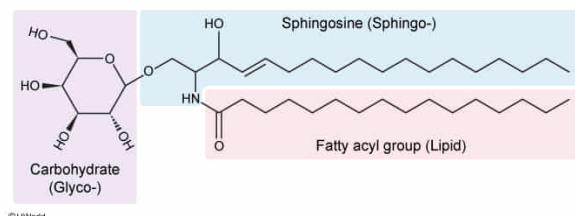
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GC can be classified as:

- ☐ A. a glycerophospholipid. [8%]
- ☐ B. a phosphosphingolipid. [2%]
- ✔ ☒ C. a glycosphingolipid. [85%]
- ☐ D. a phosphoglycolipid. [2%]

Explanation:



Lipids are a broad class of hydrophobic biomolecules that are categorized at multiple levels. At a general level, they are classified as **hydrolyzable** or **nonhydrolyzable** based on whether they contain hydrolyzable linkages such as **esters** or **amides**.

Hydrolyzable lipids are further classified based on their backbone structure and modifications. For example, lipids built on a **glycerol backbone** are glycerolipids whereas those built on **sphingosine backbones** are sphingolipids. A lipid with a phosphate group modification is called a **phospholipid**, and a lipid with a carbohydrate attached is called a **glycolipid**. Multiple characteristics can be present at once, yielding more complex names such as glycerophospholipids, phosphosphingolipids, glycosphingolipids, or phosphoglycolipids.

The passage states that GC is a hydrolyzable lipid. The structure in Figure 1 shows that it is built on a sphingosine backbone, making it a sphingolipid. Figure 1 and the passage also show that GC includes the carbohydrate galactose attached to the lipid by a glycosidic bond (glycosylation). Because GC is a sphingolipid that has been glycosylated, it can be classified as a **glycosphingolipid**.

(Choice A) Glycerophospholipids contain a phosphate group and are built on a glycerol backbone. Neither is true of GC, so it cannot be classified as a glycerophospholipid.

(Choices B and D) Phospholipids such as phosphosphingolipid or phosphoglycolipid must contain a phosphate group. However, GC does *not* contain a phosphate group, so it cannot be classified as either a phosphosphingolipid or a phosphoglycolipid.

Educational objective:

Hydrolyzable lipids are classified according to their modifications. Those with sphingosine backbones are sphingolipids; those with glycerol backbones are glycerolipids; phosphate modifications yield phospholipids; and carbohydrate modifications yield glycolipids. Complex lipids may involve a combination of these characteristics and are named accordingly (ie, glycosphingolipid, glycerophospholipid).

Time spent: 0 sec

Subject:

Biochemistry

QID: 401934

System:

Carbohydrates, Nucleotides, and Lipids